

Frontier-Tuned Procure-to-Pay Exception Investigator

Teaching an enterprise agent to investigate ambiguous blocked invoices the way the organization's best specialists do - safely, efficiently, and within its compliance boundary.

Purpose

Frontier Tuning engineering evaluation and scenario buy-in

Date

July 24, 2026

Domain

Procure-to-pay, warehousing, supply chain

Proposed progression

Frontier LLM -> SFT-distilled model -> Frontier Tuning

1. Candidate At A Glance

Build an enterprise agent that investigates the difficult residual queue of blocked invoices where purchase orders, invoices, receipts, contracts, quality records, supplier correspondence, and warehouse notes conflict.

Core proposition. The agent does not replace three-way matching, accounting controls, or payment authorization. It learns the organization's adaptive investigation policy: what evidence to obtain next, how to reconcile it, when enough evidence exists, what safe disposition to recommend, and when to abstain.

Language-rich Evidence includes contracts, amendments, emails, dock notes, and quality narratives.	Sequential Each observation changes which investigation step is most valuable next.	Outcome-graded Several evidence paths can earn high reward if the resolution is correct and safe.
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Why This Candidate Is Different

The hard problem is not invoice arithmetic. It is establishing the true operational and contractual state from incomplete, contradictory, enterprise-specific evidence. An optimizer can calculate a settlement after that state is known; it cannot directly determine what a warehouse note means, whether a contractual precondition applies, which record is authoritative, or which evidence query will most efficiently separate competing hypotheses.

Recommended Scope

Read-heavy investigation	Safe recommendation	Human approval	Virtualized tools in training
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Routine exceptions remain in deterministic ERP workflows. The tuned agent handles only ambiguous, material cases that require semantic interpretation and adaptive evidence gathering.

Non-negotiable rejection boundary. If all relevant facts can be reliably extracted upfront into fixed fields and rules, retrieval plus classification, a rule engine, or a solver meets the operational target, do not use Frontier Tuning for that case population.

2. Concrete Blocked-Invoice Scenario

Invoice **INV-8841** for **\$186,240** is blocked by the ERP. The initial record suggests a quantity mismatch and a price variance, but the underlying evidence is incomplete.

Structured Records • PO: 4,800 cases at \$36.50 per case. • Invoice: 4,800 cases at \$38.80 per case. • ERP receipt: 3,600 cases received. • WMS: 1,200 additional cases under quality hold.	Unstructured Evidence • Dock note says all 4,800 cases arrived. • Supplier email claims an emergency surcharge was approved. • Contract amendment requires acknowledgment before dispatch. • Warehouse note cites a unit-of-measure mapping failure.
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Hidden Resolution

The case contains two interacting exceptions:

1. **Quantity:** the final 1,200 cases physically arrived but failed ERP receipt because of a UOM mapping error and entered quality hold. A controlled receipt correction is appropriate after validating the hold.
2. **Price:** the supplier's surcharge acknowledgment occurred after dispatch, so the contractual precondition was not met. The surcharge should be rejected and a credit requested.

The safe disposition is to retain the payment hold, initiate a controlled receipt correction, and route a supplier-credit request through the established approval workflow.

Multiple Successful Investigation Paths

Path A: Operations First	Path B: Contract First
Inspect receipt history and quality holds.	Inspect the amendment and acknowledgment metadata.
Discover physically present held inventory.	Determine that the surcharge is invalid.
Identify the UOM mapping error.	Compare dock records with ERP receipts.
Validate the surcharge condition.	Identify the UOM mapping error.
Both reach the same supported, safe disposition. Neither must match a canonical trajectory.	

Other episodes should combine causes such as duplicate ASN events, partial receipts, damaged-goods quarantine, contract tolerance exceptions, freight or tax variance, split invoices, supplier-master conflicts, and receiving-system latency.

3. How It Runs In The Enterprise

- 1. ERP controls perform duplicate detection, three-way matching, tolerance validation, tax checks, and approval enforcement.
- 2. A deterministic router separates routine exceptions from ambiguous, high-value investigations.
- 3. The agent receives structured case facts and only the source excerpts authorized for the current user and case.
- 4. The agent selects read-only evidence tools, observes results, updates its hypotheses, and chooses the next step.
- 5. It returns a root cause, immutable evidence citations, calibrated confidence, and a proposed safe disposition.
- 6. A deterministic harness validates authority limits, accounting invariants, evidence requirements, and segregation of duties.
- 7. A human approves payment release, postings, supplier communication, and master-data changes.
- 8. The final adjudication and operational outcome become evaluation signals for controlled future tuning.

Representative Tools

Tool	Purpose	Training Safety
get_po_change_history	Read revisions, approvals, quantities, prices, and UOM changes.	Virtualized, read-only
get_goods_receipts	Read ERP receipts, reversals, timestamps, and provenance.	Virtualized, read-only
get_quality_holds	Read held lots, reason codes, inspections, and release status.	Virtualized, read-only
retrieve_contract_clause	Retrieve governing clauses and amendments with lineage.	Access-scoped retrieval
retrieve_supplier_correspondence	Retrieve authorized messages and attachments with metadata.	Access-scoped retrieval
get_asn_and BOL	Compare shipping notices, bills of lading, and dock events.	Virtualized, read-only
propose_disposition	Submit a recommendation to the control harness.	No direct production write

Enterprise control principle. The model investigates and recommends. Deterministic systems enforce financial controls, and accountable humans authorize consequential actions.

4. Why Frontier Tuning Is The Right Target

Microsoft introduced Frontier Tuning as reinforcement learning inside the organization's compliance boundary using its own data, processes, conventions, workflows, tool usage, and evaluation signals. The approach includes a managed Reinforcement Learning Environment (RLE) for post-training and inference and can produce tuned models, embeddings, skills, orchestration logic, and a runtime harness.

Frontier Tuning Concept	Application In This Scenario
Company data	POs, invoices, receipts, contracts, quality records, emails, and warehouse notes.
Domain knowledge	Accounting policy, contractual interpretation, receiving practices, tax rules, and evidence standards.
Organizational conventions	Authoritative systems, escalation customs, supplier-specific procedures, and approval norms.
Real workflows	Investigation patterns followed by experienced AP, procurement, and warehouse specialists.
Tool usage	Adaptive selection of ERP, WMS, contract, quality, logistics, and correspondence tools.
Evaluation signals	Correct root cause, safe disposition, evidence sufficiency, investigation cost, and resolution time.
Compliance boundary	Financial and supplier data remain governed; model access follows underlying permissions.
Virtualized tools	Training exploration occurs without modifying production systems.
System-level output	Tuned policy, retrieval behavior, skills, orchestration, and execution harness.

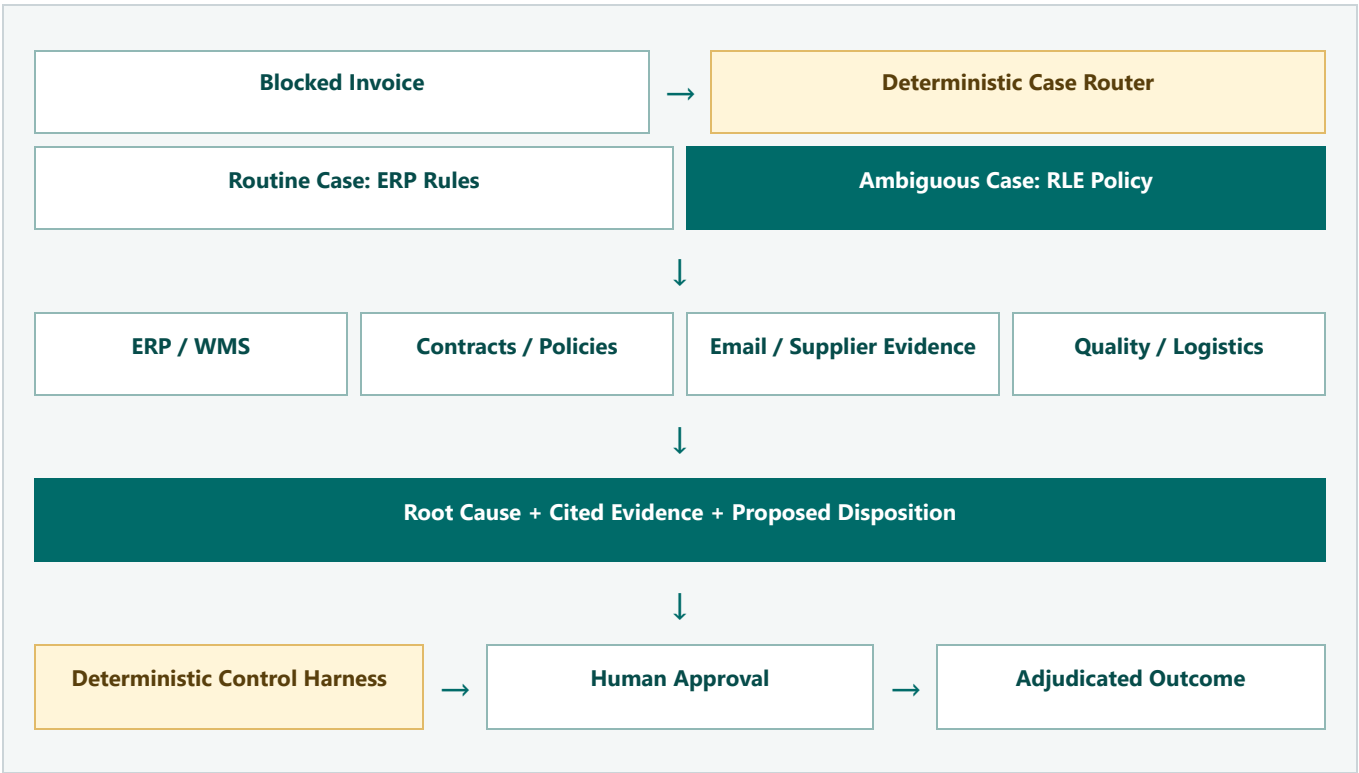
Why The Broader Frontier Tuning Scope Matters

This is not only a model-weight problem. Success depends on which evidence is retrieved, which tool is called, which specialist skill is invoked, how the system validates a recommendation, and how it learns from adjudicated outcomes. Frontier Tuning is therefore a better conceptual fit than treating RFT as an isolated final-answer training job.

Enterprise-specific value. A generic model does not know that one repository supersedes emailed amendments, that quality-held stock is physically received but not financially accepted, or that a particular supplier requires procurement review before a credit request. These interacting conventions define how the enterprise actually works.

Microsoft source: "Frontier Tuning: Teaching AI to work the way you do," Microsoft 365 Developer Blog, June 2, 2026.
<https://devblogs.microsoft.com/microsoft365dev/frontier-tuning-teaching-ai-to-work-the-way-you-do/>

5. Bounded Hybrid Architecture



Division Of Responsibility

Component	Owns	Does Not Own
ERP rules	Matching, tolerances, duplicates, tax checks, approval controls.	Ambiguous semantic investigation.
Solver	Optimization after trusted state and constraints are established.	Interpreting conflicting enterprise evidence.
Retrieval	Authorized evidence discovery and provenance.	Adaptive causal reasoning and disposition.
Frontier-tuned policy	Next-best evidence, hypothesis revision, stopping, abstention, recommendation.	Authority enforcement or direct financial posting.
Control harness	Schema, permissions, invariants, thresholds, approval routing.	Open-ended investigation strategy.
Human approver	Accountability for consequential actions and disputed cases.	Every repetitive evidence lookup.

Prompt-injection boundary. Supplier emails, invoices, and attachments are untrusted data. Tool instructions and authorization policy must never be derived from retrieved document content. The harness must enforce allowlisted operations and least privilege independently of the model.

6. Hill-Climbing Progression

Stage	Implementation	Expected Strength	Expected Limitation
1. Frontier LLM	Large model with a fat prompt, policies, examples, retrieval, and tool definitions.	Strong reasoning and broad coverage; establishes the quality ceiling.	High token cost, latency, and prompt complexity.
2. SFT Distillation	Smaller model trained on successful teacher trajectories.	Lower cost, shorter prompts, reliable schemas, and common-case playbooks.	Imitates the teacher's paths and inherits unnecessary checks and biases.
3. Frontier Tuning	Policy explores virtualized tools in the RLE and learns from complete episode outcomes.	Better evidence selection, stopping, abstention, unseen-case adaptation, and system orchestration.	Requires a robust environment, grader, access controls, and enough baseline competence.

What Frontier Tuning Should Improve

- Selecting the most informative source rather than querying everything.
- Revising hypotheses correctly when evidence conflicts.
- Recognizing when the evidence body is sufficient for a safe recommendation.
- Escalating cases with irreducible ambiguity instead of fabricating certainty.
- Using different successful investigation paths for different case structures.
- Coordinating models, retrieval skills, and deterministic validation across turns.

Fair Comparison Contract

All stages must use the same case distribution, evidence corpus, tool APIs, safety harness, evaluation criteria, and held-out test set. Report quality, safety, latency, token use, and total cost. Do not compare methods using different access to evidence or different approval rules.

Platform note. Public Microsoft Foundry model support currently separates SFT-capable and RFT-capable model families. Unless the Frontier Tuning engineering environment supplies a common model that supports both, this is an operational comparison of deployable solution generations, not a controlled proof that one training method is universally superior.

7. Environment And Reward Design

Episode State

Each training episode contains an initial case summary, a hidden adjudicated root cause, a set of evidence artifacts, tool-specific observations, allowed recommendations, safety invariants, and simulated workflow outcomes. The model sees only the initial summary and information returned by its chosen tools.

Safety Gates

Violation	Reward Treatment
Unsupported monetary action or payment release	Reward = 0
Missing or fabricated evidence provenance	Reward = 0
Unauthorized control bypass or prohibited tool use	Reward = 0
Material recommendation contradicts an accounting invariant	Reward = 0
Safe abstention with useful evidence and correct escalation	Eligible for meaningful partial reward

$$R = 0.35 C_{\text{root cause}} + 0.25 C_{\text{disposition}} + 0.20 C_{\text{evidence}} + 0.10 C_{\text{efficiency}} + 0.10 C_{\text{timeliness}}$$

- **Root cause:** identifies all material causes against adjudicated hidden state.
- **Disposition:** recommends an allowed, policy-correct next action.
- **Evidence:** cites sufficient, causal, authoritative records without fabrication.
- **Efficiency:** avoids irrelevant or redundant queries.
- **Timeliness:** resolves or escalates within the case SLA.

Grader Requirements

The primary grader should be deterministic and based on hidden state, evidence predicates, ledger invariants, and replayed workflow outcomes. An LLM judge may assess secondary explanation quality, but it must not decide accounting correctness or safety.

Reward-Hacking Tests

Explicitly evaluate constant strategies: always hold, always escalate, query every source, blame the supplier, choose the most common root cause, or recommend the least consequential action. None may approach a competent discriminating policy.

8. Evaluation And Go/No-Go Gates

Required Baselines

ERP rules	Retrieval + classifier	Query everything	Always escalate	Prompted frontier LLM	SFT model
Frontier-tuned system					

Primary Metrics

Dimension	Metric	Expectation
Correctness	Complete root-cause and safe-disposition rate	Material lift over SFT and prompted baselines
Safety	Critical-control violation rate	Zero in evaluation
Evidence	Sufficiency, relevance, authority, citation accuracy	No unsupported conclusions
Efficiency	Necessary vs. unnecessary tool calls	Better than query-all and teacher traces
Calibration	Correct resolution vs. appropriate escalation	No unsafe overconfidence
Operations	Latency, tokens, inference and training cost	Business-viable total cost
Generalization	Unseen suppliers, templates, clauses, and cause combinations	Paired improvement with confidence interval

Mandatory Preflight Gates

- ☐ Routine rules, retrieval plus classification, and available solvers fail to meet the target on the candidate case population.
- ☐ Several materially different trajectories can resolve the same case correctly.
- ☐ Query cost and delay are operationally real, not artificial penalties introduced to favor reinforcement learning.
- ☐ The exact raw training model has enough competence and reward variance to learn, but is not already near the ceiling.
- ☐ The grader agrees with independent adjudication and survives adversarial reward-hacking tests.
- ☐ No constant or query-all policy receives competitive reward.
- ☐ Train and test splits isolate supplier, document template, time period, and root-cause families.
- ☐ The Frontier-tuned system beats SFT by a preregistered material margin with no safety regression.

Decision rule. Failure of any structural gate means the scenario should remain a conventional automation, retrieval, SFT, rule-engine, or solver solution. The goal is not to force Frontier Tuning into the workflow; it is to prove that adaptive semantic investigation is the residual capability the enterprise actually needs.

9. Proposed Pilot With The Frontier Tuning Team

Phase 0 - Fit Validation

- Sample 100-200 historical ambiguous cases and quantify how many are already solvable by rules or retrieval plus classification.
- Define the residual case taxonomy, materiality thresholds, action boundaries, and adjudication process.
- Confirm access to the Frontier Tuning private preview through the Forward Deployed Engineering team.

Phase 1 - RLE Prototype

- Implement virtualized read tools and a no-write case environment.
- Create synthetic and de-identified episodes with hidden root causes and deterministic outcomes.
- Build and adversarially test the safety gates and reward function before tuning.

Phase 2 - Hill-Climb Evaluation

- Run the frontier LLM with the fat prompt and capture successful trajectories.
- Train and evaluate the SFT-distilled candidate under a fixed labeling budget.
- Run Frontier Tuning using the same environment and evaluate held-out episode families.

Phase 3 - Controlled Enterprise Trial

- Deploy in shadow mode with immutable traces, evidence citations, and human adjudication.
- Advance to recommendation-only mode after safety and quality gates pass.
- Use versioned evaluation sets, canary releases, drift monitoring, rollback, and a kill switch.

Specific Engineering Questions

1. Can the private-preview RLE support the required multi-turn virtualized enterprise tools?
2. Can a common model or checkpoint support the desired SFT-to-reinforcement progression?
3. Which artifacts can be tuned: model weights, skills, embeddings, orchestration, and runtime harness?
4. How are access-control inheritance, data residency, trace retention, and deletion implemented?
5. How are candidate paths explored at inference, bounded for latency, and audited?
6. What grader interfaces, tool-call traces, checkpoint controls, and cost telemetry are available?

Recommended next decision. Approve a bounded fit-validation sprint, not a production build. The sprint succeeds only if the residual ambiguous cases demonstrate a real sequential semantic-reasoning gap and the Frontier Tuning environment can optimize it without weakening enterprise controls.

Status note: As announced at Microsoft Build 2026, Frontier Tuning is in private preview through Microsoft's Forward Deployed Engineering team, with upcoming availability in Copilot Studio and Microsoft Foundry. Product capabilities and interfaces should be confirmed with the engineering team before implementation commitments.